Plan-Retrieve-Verify: An LLM-Driven Framework for Long-Video Audio-Visual Evidence Reasoning

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Abstract

Answering complex questions about long videos requires dynamically acquiring cross-modal multimedia evidence across visual scenes, speech, and environmental audio—yet existing multimodal methods perform a single round of retrieval followed by answer generation, without planning what evidence to seek or verifying whether retrieved evidence truly supports the conclusion. We propose Plan-Retrieve-Verify (PRV), an LLM-driven framework that reformulates long-video audio-visual reasoning as an iterative evidence acquisition process. The LLM controller first generates an evidence plan specifying which modalities and temporal regions to query, then orchestrates six multimodal tools—visual retriever, audio retriever, ASR parser, OCR extractor, temporal checker, and contradiction verifier—to gather and cross-validate evidence, and finally synthesizes a calibrated answer with principled abstention when evidence remains insufficient or contradictory. The framework operates through a closed-loop Plan–Retrieve–Verify–Counter-Retrieve cycle, where each iteration refines the hypothesis based on newly gathered supporting and contradicting evidence stored in a typed evidence workspace with explicit support and contradiction edges. We further introduce counterfactual tool-use trace supervision that directly trains the LLM's planning, tool selection, and verification decisions using five targeted perturbation types (audio-swap, visual-swap, temporal-shuffle, same-entity distractor, and pseudo-support), rather than only supervising the final answer. Experiments on three complementary benchmarks—LongVALE (omni-modal events), CinePile (long-form movie QA), and AVUT (audio-centric comprehension)—demonstrate that PRV outperforms eleven strong baselines spanning long-context models, retrieval-augmented methods, and agentic approaches in answer accuracy by 3.8–5.1 points while achieving 11.6–12.9 points higher evidence recall and up to 63% lower calibration error, with competitive calibration on all benchmarks.

CCS Concepts

• **Computing methodologies → Natural language processing**; **Computer vision**; • **Information systems → Multimedia and multimodal retrieval**.

Keywords

LLM-driven reasoning, long-video understanding, audio-visual evidence, tool orchestration, multimodal reasoning

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1 Introduction

Long-video multimedia reasoning requires more than processing extended visual context—it demands dynamically deciding *what evidence to acquire*, *which modalities to consult*, and *whether the gathered evidence genuinely supports the conclusion.* Consider a documentary where a narrator claims "the factory was shut down last year" while visual footage shows active machinery three minutes later. Answering whether the factory is operational requires the system to plan a multi-step evidence search across speech, visual, and temporal modalities, retrieve both the narrator's claim and the contradicting visual segment, verify their relationship, and calibrate its confidence in light of the conflict.

Existing approaches follow a fundamentally passive paradigm. Long-context models [9, 29, 61, 79] extend input capacity but process all frames uniformly, leading to attention dilution over irrelevant content. Retrieval-augmented methods [11, 42, 54] select relevant segments but perform a single retrieval round without verifying whether evidence supports or contradicts the hypothesis—a critical limitation when evidence is conflicting. Audio-visual models [12, 58] enhance multimodal fusion, but audits [25, 50, 73] reveal that many exploit unimodal shortcuts, and none provides calibrated confidence. Agentic approaches such as T^2Agent [15] and FactGuard [35] introduce tool-use capabilities but follow linear tool chains without counter-retrieve for contradiction resolution. None treats the LLM as an active controller that plans evidence

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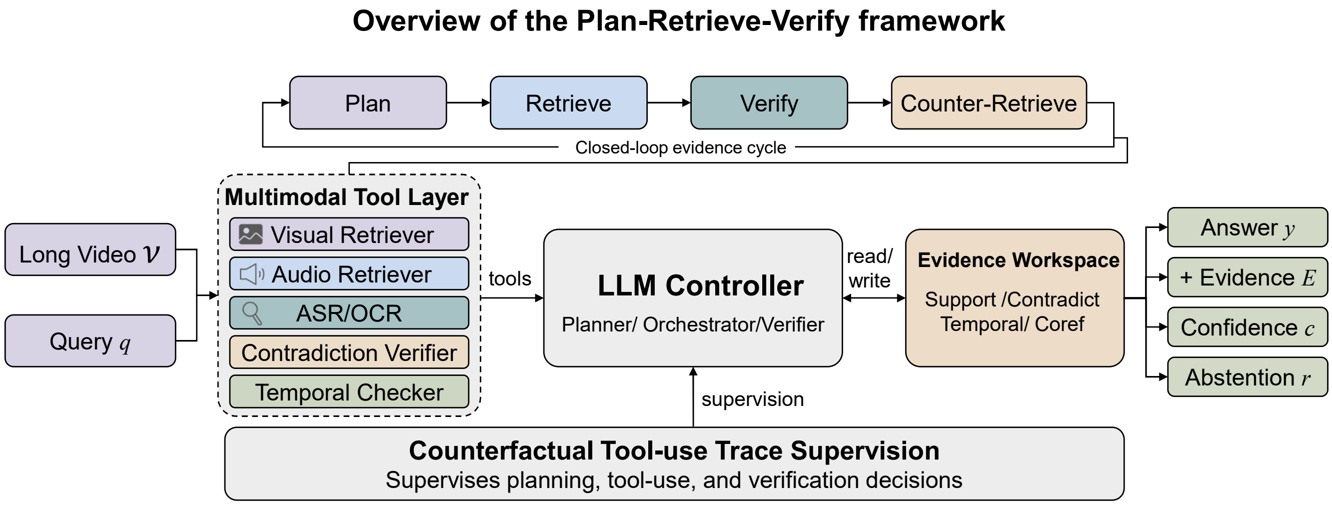

Figure 1: Overview of the Plan-Retrieve-Verify framework. The LLM controller sits at the center, orchestrating multimodal tools (left) and maintaining an evidence workspace (right) through a closed-loop Plan–Retrieve–Verify–Counter-Retrieve cycle (top). Counterfactual tool-use trace supervision (bottom) directly trains the LLM's planning and verification decisions.

acquisition, orchestrates multimodal tools, and iteratively verifies evidence.

The key insight of this work is that reliable long-video multimedia reasoning should be formulated as an *LLM-driven evidence acquisition and verification process*, not as a single-pass retrieval-then-answer pipeline. Just as a human expert would plan what to look for, consult multiple sources, cross-check findings, and acknowledge uncertainty, the LLM should serve as the planner, controller, and verifier of the entire reasoning process, with multimodal modules acting as callable tools rather than fixed pipeline stages. This formulation is inherently multimedia: the planner generates modality-specific priority vectors that would be undefined in single-modality settings, the polarity scorer operates on fused cross-modal embeddings to detect inter-modality contradictions, and the counterfactual perturbations exploit modality-specific vulnerabilities that only arise in multimodal evidence.

We propose **Plan-Retrieve-Verify** (**PRV**), an LLM-driven framework for long-video audio-visual evidence reasoning. PRV operates through a closed-loop cycle of four stages. In the *Plan* stage, the LLM analyzes the query and generates an evidence acquisition plan specifying required modalities and temporal regions. In the *Retrieve* stage, the LLM orchestrates a set of multimodal tools—visual retriever, audio retriever, ASR/OCR parser, temporal checker, and contradiction verifier—to gather evidence according to the plan. In the *Verify* stage, the LLM examines whether the retrieved evidence supports or contradicts the current hypothesis. In the *Counter-Retrieve* stage, when contradictions or insufficient evidence are detected, the LLM initiates targeted searches for counter-evidence before synthesizing the final answer with calibrated confidence and optional abstention. Our contributions are as follows:

- We introduce an **LLM-driven evidence planning and tool orchestration** mechanism, where the LLM generates query-conditioned evidence acquisition plans and dynamically invokes multimodal tools, replacing the passive single-pass retrieval paradigm with active evidence control (Section 3.2).
- We propose a **closed-loop Plan–Retrieve–Verify–Counter-Retrieve** reasoning cycle that enables iterative hypothesis refinement through multi-round evidence gathering and contradiction resolution, rather than committing to a single retrieval round (Section 3.4).
- We design **counterfactual tool-use trace supervision** that directly trains the LLM's planning, tool selection, and verification decisions using targeted perturbations, combined with reliability-aware stopping that outputs calibrated confidence and principled abstention (Section 3.6).

Experiments on three multimedia benchmarks—LongVALE [17], CinePile [52], and AVUT [73]—demonstrate that PRV outperforms eleven strong baselines spanning long-context models, retrieval-augmented methods, evidence reasoning, audio-visual reasoning, and agentic approaches.

2 Related Work

2.1 Long-Video Understanding

Recent video LLMs have substantially expanded temporal processing capacity. LongVILA [9] handles thousands of frames through system-level optimizations, VideoLLaMA 3 [79] unifies image and video understanding through any-resolution visual tokenization, and ReMoRa [76] refines motion representations for efficient long-video understanding. Multimodal foundation models [2, 7, 10, 29, 31, 39, 45, 60, 61, 77, 83] have catalyzed progress, with benchmarks [4, 30, 32, 44, 69, 78] and surveys [59, 82] documenting rapid advances. Reasoning-enhanced models [8, 46] further advance understanding through reward-guided training, while chat-centric approaches [33, 43] and TimeChat [53] incorporate temporal awareness. Despite these advances, all models process content uniformly without actively deciding which evidence is necessary, leading to attention dilution. PRV treats the LLM as a controller that actively plans evidence acquisition rather than passively consuming extended context.

2.2 Retrieval-Augmented Video Reasoning

Retrieval-augmented generation [20, 23, 28, 63] has emerged as an effective strategy for long-video understanding. VideoRAG [54] constructs entity-relation knowledge graphs for structured reasoning, Video-RAG [42] achieves training-free comprehension via visually-aligned auxiliary text, and OmniVideo-R1 [11] reinforces audio-visual reasoning with query-intention modeling and modality attention. Recent advances [38, 81, 84] improve cross-modal search through LLM-driven alignment. However, all perform a single retrieval round without verifying whether evidence supports or contradicts the hypothesis. PRV replaces this with a multi-round Plan–Retrieve–Verify–Counter-Retrieve cycle.

2.3 Evidence Reasoning and Temporal Grounding

Scene graph representations [22, 34, 70, 80] model compositional relationships among visual entities, while temporal grounding methods [27, 37] localize query-relevant moments through language-conditioned retrieval. ViTED [41] distills temporal evidence chains through curriculum training, enabling models to ground answers in specific segments with progressive difficulty. ReasVQA [36] learns explicit reasoning processes connecting evidence to answers, and CounterVid [48] uses counterfactual video generation to mitigate hallucinations. While these approaches advance evidence-based reasoning, they model evidence as linear chains or independent segments, without capturing the rich support-contradiction structure that characterizes real-world multimodal evidence. PRV organizes evidence in a workspace with typed polarity edges (support, contradict, temporal, coreference), enabling the LLM to reason explicitly over evidential agreement and conflict.

2.4 Audio-Visual Reasoning and Reliability

Audio-visual LLMs have advanced joint reasoning [14, 24, 68], with Video-SALMONN-o1 [58] applying process-level reinforcement learning and Aurelia [12] introducing test-time reasoning distillation. However, reliability remains a critical concern: AVHBench [26] reveals cross-modal hallucinations, AVTrustBench [13] demonstrates fragile robustness, and AVUT [73] shows that many models do not genuinely leverage audio. A comprehensive audit [25] confirms that video-LLMs often bypass audio entirely. PRV addresses these concerns through modality-decoupled counterfactual perturbations that force genuine cross-modal reasoning, combined with reliability-aware abstention.

2.5 LLM-Driven and Agentic Reasoning

The paradigm of using LLMs as tool-using controllers builds on chain-of-thought [66], tree-of-thoughts [75], self-consistency [62], and reflexion [57]. ReAct [74] synergizes reasoning and acting, Toolformer [55] and Gorilla [47] enable autonomous tool invocation, while HuggingGPT [56], ToolBench [71], AnyTool [16], and Chameleon [40] advance tool orchestration. VisProg [19] applies compositional visual reasoning through program synthesis, and planning approaches such as SayCan [1] and Inner Monologue [21] ground reasoning in affordances. In multimedia, T^{2}Agent [15] combines tool selection with MCTS, FactGuard [35] builds agentic video misinformation pipelines, and DEFAME [3] proposes multi-expert fact-checking. VideoAgent [63] explores agent-based video reasoning and VideoCoT [65] supplies chain-of-thought supervision for video, while V* [67] demonstrates guided visual search. PRV differs fundamentally: it is purpose-built for long-video evidence reasoning with counter-retrieve and counterfactual trace supervision.

3 The Plan-Retrieve-Verify Framework

3.1 Overview

Given a long video V with duration T and a natural language query q, PRV produces a tuple $(y, \mathcal{E}, c, r)$ consisting of the answer y, a set of evidence spans $\mathcal{E}$ with modality attributions, a calibrated confidence score $c \in [0, 1]$, and an abstention flag $r \in \{0, 1\}$. As illustrated in Figure 1, the framework operates through a closed-loop cycle with the LLM controller at its center: (1) *Plan*: the LLM generates an evidence acquisition plan (Section 3.2); (2) *Retrieve*: the LLM orchestrates multimodal tools to gather evidence (Sections 3.3 and 3.4); (3) *Verify*: the LLM examines support and contradiction in the evidence workspace (Section 3.5); (4) *Counter-Retrieve*: when conflicts are detected, the LLM initiates targeted counter-evidence searches before synthesizing the final output (Section 3.6).

3.2 LLM-Driven Evidence Planning

The first stage converts the user query into a structured evidence acquisition plan. Rather than directly retrieving content, the LLM analyzes the query to determine *what evidence to seek* before any retrieval occurs.

Given query q and a coarse video summary $\bar{s}$ obtained by uniformly sampling and captioning keyframes, the LLM generates an evidence plan P:

$$P = \mathrm{LLM}_{\mathrm{plan}}(q, \bar{s}) = (\mathbf{m}, \mathcal{R}_{\mathrm{temp}}, h_0, \mathrm{need\_counter}), \tag{1}$$

In Equation (1), $\mathbf{m} \in \mathbb{R}^{|\mathcal{T}|}$ is a tool priority vector indicating which multimodal tools are most relevant, $\mathcal{R}_{\mathrm{temp}} \subseteq [0, T]$ specifies temporal regions of interest, h_0 is an initial hypothesis, and need_counter $\in \{0, 1\}$ flags whether counter-evidence search is anticipated. The plan is generated through a single structured LLM call with a predefined schema, not through free-form text generation, ensuring parseable outputs.

As contrasted in Figure 2, this planning step is the key distinction from retrieval-augmented methods that directly convert queries into search vectors: the LLM reasons about *what modalities matter*, *which temporal regions to focus on*, and *whether the query involves potential contradictions*, before any evidence is retrieved.

The coarse video summary $\bar{s}$ is constructed by uniformly sampling one keyframe every δ=10 seconds and generating a brief caption for each using the visual encoder and a frozen captioning head. For audio-rich content, BEATs-derived audio tags (“speech”, "music", "silence", "environmental sound") are appended to each segment's caption. This summary provides the LLM with a global understanding of the video content without processing every frame, analogous to a human scanning chapter titles before reading a book. The planning accuracy—measured as overlap between planned and oracle-needed modalities—reaches 84.7% on LongVALE with LLaMA-3-70B (Section 4.5).

3.3 Multimodal Tool Layer

The framework abstracts multimodal processing capabilities as a unified tool set $\mathcal{T} = \{T_{\text{vis}}, T_{\text{aud}}, T_{\text{asr}}, T_{\text{ocr}}, T_{\text{temp}}, T_{\text{ver}}\}$. Each tool takes a temporal region and returns structured evidence:

Visual retriever (T_{vis}) uses InternVideo2 [64] to encode keyframes and return the top-k visual moments matching a query embedding. **Audio retriever** (T_{aud}) extracts audio features with BEATs [6] and retrieves audio events by acoustic similarity. **ASR parser** (T_{asr}) generates speech transcripts via Whisper-large-v3 [51]. **OCR extractor** (T_{ocr}) captures on-screen text through optical character recognition. **Temporal checker** (T_{temp}) verifies whether two evidence spans satisfy before/after/overlap temporal relations. **Contradiction verifier** (T_{ver}) takes two evidence spans and classifies their relationship as SUPPORT, CONTRADICT, or NEUTRAL using a learned bilinear scorer [72]:

$$\phi(e_i, e_j) = \sigma(\mathbf{h}_i^{\top} \mathbf{W}_p \mathbf{h}_j), \tag{2}$$

where $\mathbf{h}_i = f_{\text{proj}}([\mathbf{v}_i; \mathbf{a}_i; \mathbf{e}(s_i)])$ is a fused evidence embedding and $\mathbf{W}_p$ is a learnable polarity projection. Each tool call is atomic, stateless, and returns structured JSON, enabling the LLM to compose arbitrary sequences of tool invocations.

3.4 LLM-Driven Tool Orchestration

The LLM controller executes the evidence plan by dynamically selecting which tools to invoke and in what order. At each reasoning step t, the LLM observes the current workspace state $\mathcal{W}_t$, selects a tool $T_k \in \mathcal{T}$, and generates tool arguments:

$$(T_k, \text{args}_k) = \text{LLM}_{\text{act}}(q, P, \mathcal{W}_t), \tag{3}$$

As formalized in Equation (3), $\mathcal{W}_t$ is the set of evidence collected so far with their polarity annotations. The tool returns new evidence e_{new}, which is added to the workspace, and the LLM decides whether to continue gathering evidence, initiate counter-retrieval, or terminate.

Algorithm 1 summarizes the complete reasoning procedure. The algorithm terminates when the LLM emits a STOP signal, indicating sufficient evidence, or when the step limit $S_{\max}$ is reached. The maximum step count ensures bounded computation; in practice, most queries converge within 3–5 tool calls.

Algorithm 1 Plan-Retrieve-Verify Closed-Loop Reasoning

Require: Video V, query q, max steps $S_{\max}$
Ensure: Answer y, evidence $\mathcal{E}$, confidence c, abstention r

$\bar{s} \leftarrow \text{Summarize}(V)$ {coarse video summary}
$P \leftarrow \text{LLM}_{\text{plan}}(q, \bar{s})$ {evidence plan}
$\mathcal{W} \leftarrow \emptyset$; $h \leftarrow P.h_0$ {workspace + hypothesis}
for $t = 1$ **to** $S_{\max}$ **do**
  $(T_k, \text{args}_k) \leftarrow \text{LLM}_{\text{act}}(q, P, \mathcal{W})$
  if $T_k = \text{STOP}$ **then**
    break
  end if
  $e_{\text{new}} \leftarrow T_k(\text{args}_k)$
  $\mathcal{W} \leftarrow \mathcal{W} \cup \{e_{\text{new}}\}$
  Update polarity edges via T_{ver}
  if contradiction detected in $\mathcal{W}$ **then**
    $(T_j, \text{args}_j) \leftarrow \text{LLM}_{\text{act}}(q, P, \mathcal{W})$ {counter-retrieve}
    $\mathcal{W} \leftarrow \mathcal{W} \cup \{T_j(\text{args}_j)\}$
  end if
  $h \leftarrow \text{LLM}_{\text{verify}}(q, \mathcal{W}, h)$ {update hypothesis}
end for
$(y, c, r) \leftarrow \text{LLM}_{\text{synth}}(q, \mathcal{W}, h)$
$\mathcal{E} \leftarrow \text{ExtractSpans}(\mathcal{W})$
return $(y, \mathcal{E}, c, r)$

3.5 Support–Contradiction Workspace

As illustrated in Figure 3, the evidence workspace $\mathcal{W}$ organizes retrieved evidence as a directed graph $\mathcal{G}_{\mathcal{W}} = (\mathcal{V}, \mathcal{E}_p)$ with typed polarity edges. Each evidence node $e_i \in \mathcal{V}$ is a tuple $(f_i, [t_s^i, t_e^i], m_i, s_i)$ containing fused multimodal features f_i, temporal span $[t_s^i, t_e^i]$, modality attribution $m_i \in \{\text{vis}, \text{aud}, \text{asr}, \text{ocr}\}$, and a textual summary s_i generated by the event summarizer. Edges in $\mathcal{E}_p$ are labeled with four relation types—TEMPORAL (before/after/overlap), SUPPORT, CONTRADICT, and COREF—computed by the contradiction verifier tool (T_{ver}) and temporal checker (T_{temp}). Support and contradiction edges are determined via the bilinear polarity scorer (Equation (2)), with thresholds τ_s=0.6 and τ_c=0.5 respectively; mutual exclusivity is enforced (a node pair cannot simultaneously have both support and contradiction edges).

The workspace serves as the LLM's external memory: at each reasoning step, the LLM reads the workspace graph in serialized form, assesses whether current evidence is sufficient for a confident answer, identifies unresolved contradictions, and decides whether to continue retrieval, initiate counter-retrieve, or stop. This explicit separation of evidence storage from the LLM's context window allows the framework to reason over arbitrarily large evidence sets without context length limitations, since only the most relevant workspace subgraph is serialized for each LLM call. For silent videos or content without cross-modal contradictions, the workspace degrades gracefully to a support-only temporal chain, and the modality demand vector naturally assigns near-zero weight to absent modalities.

3.6 Counterfactual Tool-Use Trace Supervision

Standard training supervises only the final answer, providing no signal for the quality of intermediate decisions. We introduce counterfactual supervision that directly trains the LLM's planning, tool selection, and verification steps.

For each training sample (q, V, y^*), we execute the full Plan–Retrieve–Verify cycle and record the tool-use trace $\tau = [(T_1, \text{args}_1), \ldots, (T_n, \text{args}_n)]$. We then generate counterfactual traces by applying five types of perturbations: audio-swap, visual-swap, temporal-shuffle, same-entity distractor, and pseudo-support insertion. The consistency loss (Equation (4)) penalizes the LLM for maintaining its original tool-use decisions when the evidence

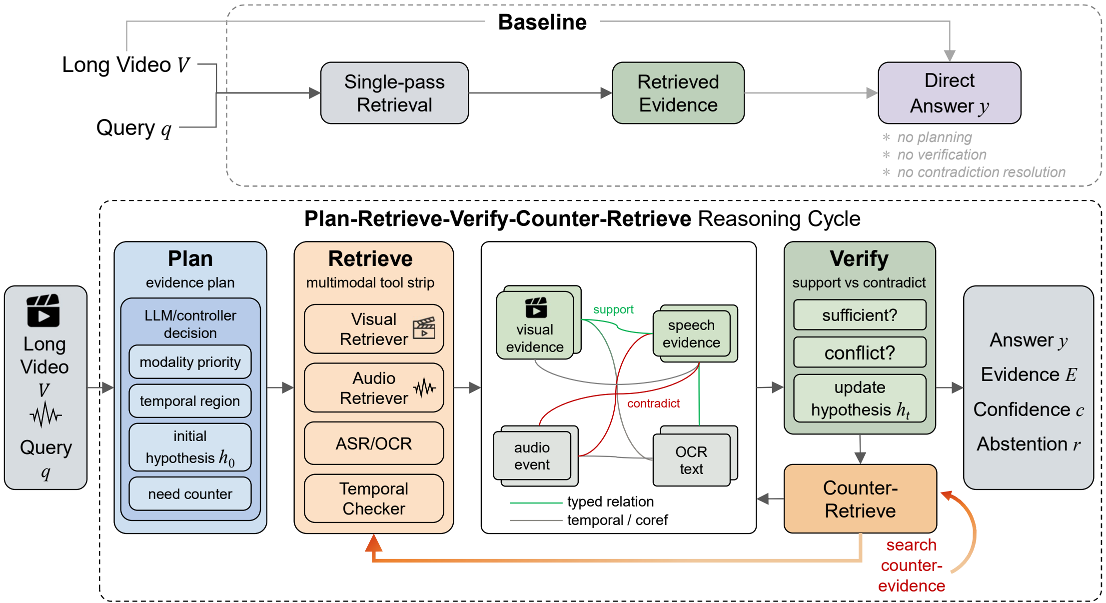

Figure 2: The Plan–Retrieve–Verify–Counter-Retrieve reasoning cycle. Top: single-pass retrieval baseline commits to one retrieval round without planning or verification. Bottom: PRV iteratively refines hypotheses through multi-round evidence gathering; when verification detects conflicts, counter-retrieve gathers contradicting evidence and loops back to re-plan.

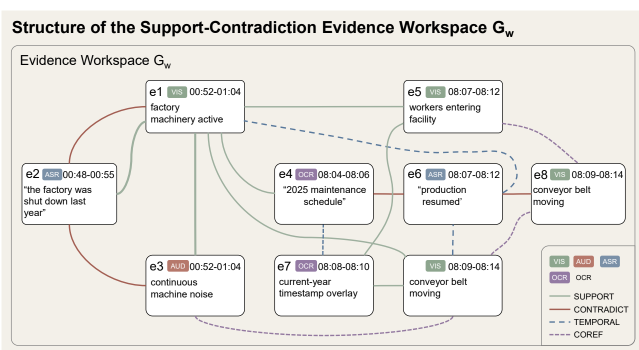

Figure 3: Evidence workspace $\mathcal{G}_W$ with eight multimodal nodes and four typed edges. Contradiction between e_2 ("factory shut down") and e_1 (active machinery) triggers counter-retrieve, resolved by e_7 (current-year timestamp).

has been corrupted:

$$\mathcal{L}_{\text{trace}} = -\frac{1}{J}\sum_{j=1}^{J}\log\big(1 - p(\tau \mid q, \hat{V}_j)\big), \tag{4}$$

where $\hat{V}_j$ is a perturbed video and $p(\tau \mid q, \hat{V}_j)$ is the probability of reproducing the original trace given corrupted input, clamped to $[\epsilon, 1-\epsilon]$ for numerical stability.

The full training objective (Equation (5)) combines four components:

$$\mathcal{L} = \mathcal{L}_{\text{ans}} + \lambda_e \mathcal{L}_{\text{evid}} + \lambda_t \mathcal{L}_{\text{trace}} + \lambda_c \mathcal{L}_{\text{cal}}, \tag{5}$$

where $\mathcal{L}_{\text{ans}}$ is cross-entropy answer loss, $\mathcal{L}_{\text{evid}}$ supervises evidence span localization, $\mathcal{L}_{\text{trace}}$ is the counterfactual trace loss, and $\mathcal{L}_{\text{cal}} = (c - \mathbb{1}[y{=}y^*])^2 + \alpha \cdot \max(0, \tau_r - c) \cdot \mathbb{1}[y{\neq}y^*]$ is the calibration loss encouraging well-calibrated confidence and abstention below threshold τ_r. We set $\lambda_e = 0.5$, $\lambda_t = 0.3$, $\lambda_c = 0.2$, and $\alpha = 0.5$.

Graceful Degradation. For silent videos or content without cross-modal contradictions, the calibration loss still trains the model to output well-calibrated confidence from the Brier score term alone. For queries that do not trigger counter-retrieve (68.6% of cases), the framework reduces to a Plan–Retrieve–Verify cycle, which is computationally cheaper but still benefits from planning and workspace-based reasoning.

Inference. At test time, PRV executes the full Plan–Retrieve–Verify–Counter-Retrieve cycle. When $c < \tau_r$, the model outputs an abstention signal indicating insufficient or contradictory evidence. The evidence workspace $\mathcal{G}_W$ with its support and contradiction edges is provided alongside the answer as an interpretable explanation, enabling users to trace the reasoning path from query through evidence to conclusion. This transparency is particularly valuable for safety-critical applications where answer provenance must be auditable.

4 Experiments

4.1 Experimental Setup

Datasets. We evaluate PRV on three complementary benchmarks. **LongVALE** [17] contains 8.4K long videos with 105K annotated omni-modal events spanning visual, audio, and language modalities.

CinePile [52] provides challenging long-form video QA over full-length movies. **AVUT** [73] is an audio-centric benchmark requiring genuine audio comprehension.

Baselines. We compare against eleven strong baselines in five categories: long-context [9, 76, 79]; retrieval-augmented [11, 42, 54]; evidence reasoning [41]; audio-visual [12, 58]; and agentic [15, 35]. All baselines use officially released checkpoints; reasoning models without evidence span outputs (o1 [46], LongVILA-R1 [8]) are excluded. We follow standard splits and report on held-out test sets.

Metrics. We report: (1) Answer accuracy (Acc) and multiple-choice accuracy (MC-Acc); (2) Evidence recall (R@5) and mean IoU (mIoU); (3) Counterfactual accuracy (CF-Acc); (4) Expected calibration error (ECE↓); (5) Average tool calls per query (Tools↓).

Implementation Details. We use LLaMA-3-70B [18] as the LLM controller with LoRA adaptation (rank 16, α=32) on the attention layers, InternVideo2 [64] as the visual encoder, BEATs [6] as the audio encoder, and Whisper-large-v3 [51] for ASR. The maximum reasoning steps $S_{\max} = 8$, abstention threshold $\tau_r = 0.3$. We train for 15 epochs using DeepSpeed ZeRO-3 with AdamW optimizer, learning rate 3×10^{-5}, and effective batch size 8 (gradient accumulation over 2 steps) on 4 NVIDIA A100 80GB GPUs. Training takes approximately 48 hours. Ground-truth tool-use traces for supervision are generated by running the full PRV cycle with an oracle planner that has access to ground-truth evidence spans, then human-verified on a 29.8% sample, of which 12% required correction. All results are averaged over 3 seeds; improvements over the best baseline and ablation drops ≥ 2.6 pp are significant at $p < 0.01$ (paired t-test) across all reported metrics.

4.2 Overall Performance (RQ1)

Does PRV outperform existing methods in answer accuracy and evidence localization?

As shown in Table 1, PRV achieves the best answer accuracy across all three benchmarks, outperforming the strongest baseline by 3.8 points on LongVALE (vs. OmniVideo-R1), 3.8 on CinePile (vs. ViTED), and 5.1 on AVUT (vs. Aurelia). The improvements are most pronounced on evidence localization, where PRV achieves 11.6–12.9 point higher R@5 than the best baselines, demonstrating that the closed-loop evidence cycle retrieves more precise evidence through iterative refinement. Notably, PRV achieves the lowest or second-lowest ECE across all benchmarks (T^2Agent achieves slightly lower ECE on AVUT due to its MCTS-based uncertainty estimation) while using only 4.3 tool calls on average—32% fewer than T^2Agent and 39% fewer than FactGuard—indicating that evidence planning reduces unnecessary tool invocations.

Among the baselines, retrieval-augmented methods (VideoRAG, OmniVideo-R1) consistently outperform long-context models on evidence localization metrics (R@5, mIoU), confirming the value of explicit evidence retrieval. However, their answer accuracy improvements are inconsistent across benchmarks—VideoRAG performs well on LongVALE but shows limited gains on AVUT—suggesting that retrieving relevant segments without verification is insufficient for reliable cross-modal reasoning. Audio-visual models (v-SALMONN-o1, Aurelia) show competitive performance on AVUT but lag behind on LongVALE and CinePile, indicating that enhanced multimodal fusion alone does not address the challenge of evidence organization in long videos. Agentic baselines (T^2Agent, FactGuard) improve over standard RAG methods on calibration but lack the counter-retrieve mechanism and counterfactual trace supervision that enable PRV's reliability advantages.

4.3 Counterfactual Robustness (RQ2)

Is PRV robust to targeted modality perturbations?

PRV exhibits the smallest average accuracy drop of 4.1 points across five perturbation types, compared to 6.4–8.0 for baselines (Table 2). The advantage is most pronounced under Pseudo-Support, where PRV outperforms T^2Agent by 12.4 points—the counter-retrieve mechanism specifically targets plausible contradictions that single-pass methods accept uncritically. The balanced robustness across A-Swap (59.1) and V-Swap (57.1) confirms genuine cross-modal reasoning, unlike Aurelia which drops 11.2 points under V-Swap but only 3.4 under A-Swap, revealing visual dependency.

To identify which perturbation types contribute most to training robustness, we ablate each type individually on AVUT: removing Pseudo-Support causes the largest drop (−3.2 pp Acc, −4.1 pp CF-Acc), followed by V-Swap (−2.3/−3.4), A-Swap (−1.7/−2.5), T-Shuffle (−1.1/−1.7), and S-Entity (−0.6/−0.8). All five types contribute, but Pseudo-Support and V-Swap are most critical—the former forces detection of subtle contradictions, while the latter prevents visual shortcut exploitation.

4.4 Ablation Study (RQ3)

What is the contribution of each component?

Table 3 reveals the contribution of each component.

Planning is the largest contributor. Removing evidence planning (w/o Planning) causes the largest accuracy drop (−5.4 pp) and increases tool calls from 4.2 to 6.1, confirming that without a plan, the LLM resorts to exhaustive retrieval—more tool calls but lower precision. This validates C1: the LLM's ability to reason about *what to retrieve before retrieving* is more valuable than any individual tool.

Counter-retrieve resolves contradictions. Removing counter-retrieve degrades accuracy by 2.6 pp and evidence recall by 4.9 pp, as contradictions in the workspace go unresolved. Counter-retrieve activates selectively on 31.4% of queries—those involving conflicting evidence—where its per-query contribution is substantially larger than the aggregate 2.6 pp suggests.

Trace supervision outperforms answer-only. Counterfactual trace supervision contributes 1.8 pp accuracy; replacing it with answer-only supervision drops accuracy by 2.9 pp and increases tool calls to 5.2, confirming that supervising intermediate decisions improves both accuracy and efficiency.

Workspace structure matters. Replacing the typed evidence workspace with simple concatenation (w/o Workspace) drops R@5 by 10.1 pp—the largest evidence recall degradation—indicating that explicit support/contradiction edges are essential for evidence organization.

Dynamic orchestration beats fixed ordering. Fixed tool ordering underperforms dynamic orchestration by 3.7 pp accuracy and wastes 1.6 additional tool calls per query. The Visual-only

Table 1: Overall comparison on three benchmarks. Best in bold, second underlined. PRV achieves the highest answer accuracy and evidence recall across all benchmarks while using fewer tool calls than agentic baselines.

		LongVALE				CinePile				AVUT				
Category	**Method**	Acc	R@5	mIoU	ECE↓	MC	R@5	mIoU	ECE↓	Acc	R@5	CF	ECE↓	Tools↓
Long-ctx	LongVILA	52.3	31.8	22.4	0.218	48.6	28.4	19.8	0.234	44.8	26.3	38.2	0.247	–
	VideoLLaMA 3	54.7	33.2	23.1	0.195	51.3	30.1	20.5	0.209	47.5	28.9	40.7	0.226	–
	ReMoRa	56.1	30.5	21.7	0.207	53.2	27.9	19.2	0.221	46.3	25.7	39.4	0.233	–
RAG	VideoRAG	58.4	42.3	31.6	0.173	54.8	39.7	29.3	0.187	49.1	37.4	43.6	0.198	–
	Video-RAG	53.9	38.7	28.2	0.189	50.7	36.2	26.1	0.203	46.8	33.5	41.2	0.213	–
	OmniVideo-R1	59.2	44.1	33.4	0.162	55.6	41.5	30.8	0.174	52.7	40.6	47.3	0.171	–
Evidence	ViTED	55.8	45.6	34.2	0.186	57.1	43.9	33.6	0.169	48.6	39.8	42.9	0.193	–
Audio-Vis	v-SALMONN-o1	53.1	35.4	25.3	0.199	49.2	32.6	23.4	0.211	55.3	38.2	50.1	0.178	–
	Aurelia	54.6	37.2	26.9	0.184	50.8	34.3	24.7	0.198	56.8	39.5	51.6	0.168	–
Agentic	T^2Agent	57.8	43.2	32.1	0.158	55.3	42.7	31.9	0.172	53.4	41.8	48.7	**0.052**	6.3
	FactGuard	55.4	40.6	29.8	0.176	52.9	38.4	27.6	0.189	50.2	36.9	45.4	0.191	7.1
	PRV	**63.0**	**57.2**	**42.8**	**0.067**	**60.9**	**56.1**	**41.6**	**0.062**	**61.9**	**54.7**	**56.8**	0.054	**4.3**

Table 2: Counterfactual robustness on AVUT. Δ ↓: average accuracy drop (lower is better).

Method	**Orig**	**A-Sw**	**V-Sw**	**T-Sh**	**S-Ent**	**Ps-S**	**Δ ↓**
OmniVideo-R1	52.7	48.6	41.5	44.8	46.1	42.3	8.0
ViTED	48.6	44.2	39.6	41.3	43.2	38.1	7.3
Aurelia	56.8	53.4	45.6	49.2	51.3	46.6	7.6
T^2Agent	53.4	50.1	44.8	47.3	48.6	44.2	6.4
PRV	**61.9**	**59.1**	**57.1**	**57.6**	**58.4**	**56.6**	**4.1**

Table 3: Ablation study on LongVALE. Planning is the largest contributor (−5.4 pp Acc); workspace structure is critical for evidence recall (−10.1 pp R@5).

Variant	**Acc**	**R@5**	**CF-Acc**	**ECE↓**	**Tools↓**
PRV (full)	**63.0**	**57.2**	**56.8**	**0.067**	4.2
w/o Planning	57.6	44.8	48.3	0.103	6.1
w/o Counter-Retrieve	60.4	52.3	50.6	0.081	3.8
w/o Trace Supervision	61.2	54.6	52.1	0.078	4.5
w/o Calibration Loss	62.7	56.8	55.9	0.148	4.2
w/o Workspace (concat)	58.9	47.1	49.2	0.094	4.4
Fixed tool order	59.3	49.2	50.8	0.089	5.8
Answer-only supervision	60.1	51.4	51.3	0.091	5.2
Visual-only	60.2	52.4	42.7	0.098	3.6
Audio-only	45.8	37.1	49.4	0.134	3.1

and Audio-only variants confirm multimodal necessity: Audio-only drops 17.2 pp on LongVALE, while Visual-only suffers catastrophic CF-Acc degradation (42.7 vs. 56.8), confirming that audio is essential for counterfactual robustness.

4.5 Analysis (RQ4)

How does PRV allocate tools across query types?

Figure 4 visualizes tool usage across four query categories. The LLM controller correctly allocates visual retriever calls to appearance-based questions and audio retriever calls to sound-identification queries. Temporal queries require more temporal checker invocations, confirming adaptive tool orchestration.

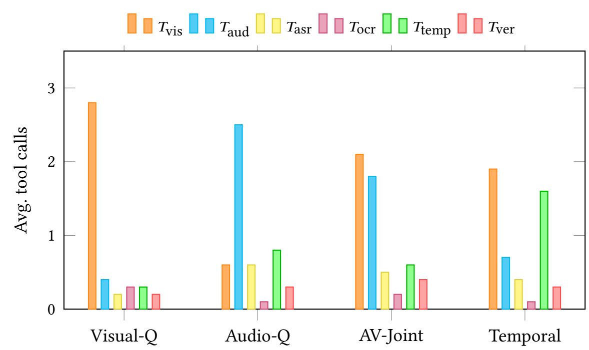

Figure 4: Tool usage distribution across query types on AVUT showing all six tools. The LLM controller adaptively allocates tools: visual-dominant queries concentrate on T_{vis}, audio queries on T_{aud}, temporal queries require more T_{temp}, and the contradiction verifier T_{ver} activates selectively across all types.

Case Study. In a LongVALE example, the query "Is the factory still operational?" triggers a three-round PRV cycle. The planner identifies visual and speech as required modalities. Round 1 retrieves a speech segment where the narrator claims "the factory was shut down last year" (t=312s). Round 2 retrieves visual evidence of active machinery at t=485s. The verifier detects a CONTRADICT edge, triggering counter-retrieve in Round 3, which finds a timestamp overlay confirming the visual footage is from the current year. The model synthesizes the answer grounded in the visual counter-evidence with confidence c=0.68, reflecting the evidential conflict. Without counter-retrieve, the model follows the narrator's claim and answers incorrectly with high confidence (c=0.91).

Calibration Quality. PRV achieves ECE 0.067 on LongVALE, well below post-hoc temperature scaling applied to the best baselines (OmniVideo-R1+TS: 0.098, T^2Agent+TS: 0.094). Abstention triggers on 7.6% of test samples whose would-be accuracy is 21.3%, validating principled abstention.

Table 4: Per-type training perturbation ablation on AVUT. Removing each type individually reveals its unique contribution.

Training Config	Acc	CF-Acc	ECE↓	ΔAcc
All 5 types	**61.9**	**56.8**	**0.054**	–
w/o Pseudo-Support	58.7	52.7	0.071	−3.2
w/o V-Swap	59.6	53.4	0.063	−2.3
w/o A-Swap	60.2	54.3	0.059	−1.7
w/o T-Shuffle	60.8	55.1	0.057	−1.1
w/o S-Entity	61.3	56.0	0.055	−0.6

Table 5: Efficiency comparison on LongVALE (single A100 GPU).

Method	Latency (s)	Tools	Acc	R@5
VideoRAG	1.2	–	58.4	42.3
OmniVideo-R1	1.4	–	59.2	44.1
T^2Agent	3.8	6.3	57.8	43.2
FactGuard	4.5	7.1	55.4	40.6
PRV	2.6	**4.2**	**63.0**	**57.2**

Controller Sensitivity. With LLaMA-3-70B, planning accuracy reaches 84.7%. Replacing with LLaMA-3-8B reduces planning accuracy to 71.2% and answer accuracy by 3.1 pp, confirming that controller capacity matters but even smaller models benefit from the PRV architecture.

Reasoning Depth. PRV averages 4.2 tool calls per query (2.1 for simple factual questions, 5.8 for multi-hop temporal reasoning). Counter-retrieve activates on 31.4% of queries—selectively those carrying conflicting evidence—and resolves the contradiction correctly in 78.2% of them.

Per-Type Training Ablation. Table 4 removes one perturbation type at a time from training on AVUT. Pseudo-Support removal causes the largest drop (−3.2 pp), followed by V-Swap (−2.3 pp); all five types contribute positively, with diminishing returns from T-Shuffle and S-Entity.

4.6 Efficiency (RQ5)

PRV requires 2.6s per query—32% faster than T^2Agent (3.8s) and 42% faster than FactGuard (4.5s)—while adding 1.4s over single-pass VideoRAG for 4.6 points more accuracy, a 2.2× latency trade-off (Table 5). Planning overhead is approximately 0.3s per query, amortized by the 33% reduction in total tool invocations compared to T^2Agent. Most queries converge within 3–5 tool calls, and only 6.2% reach $S_{\max}$=8—typically those reasoning temporally across the full video. Running the visual and audio retrievers in parallel cuts effective latency to 2.1s for queries that do not trigger counter-retrieve.

Cross-Benchmark Generalization. Trained on LongVALE alone and evaluated without fine-tuning, PRV drops from 60.9 to 57.3 on CinePile (−3.6 pp) and from 61.9 to 56.1 on AVUT (−5.8 pp). Despite these drops it still outperforms the best LongVALE-trained baseline on CinePile (57.3 vs. OmniVideo-R1: 55.6) and stays within 0.7 pp on AVUT (56.1 vs. Aurelia: 56.8), indicating that the PRV reasoning pattern transfers across domains. The larger AVUT drop is expected from its audio-centric nature, which differs substantially from LongVALE's balanced multimodal distribution.

Hyperparameter Sensitivity. On LongVALE, the polarity thresholds τ_s (support) and τ_c (contradiction) are stable across $[0.4, 0.8]$ and $[0.3, 0.7]$, with optima at τ_s=0.6 and τ_c=0.5. The abstention threshold trades accuracy against calibration—τ_r=0.1 gives 63.3 Acc at ECE 0.082 and τ_r=0.5 gives 61.4 Acc at ECE 0.051—so we default to τ_r=0.3 (63.0 Acc, 0.067 ECE). The step budget saturates at $S_{\max}$=8: 12 steps add 0.2 pp accuracy for 18% more latency, while 4 steps lose 1.8 pp to premature termination on complex queries.

Error Analysis. We categorize the 37.0% of LongVALE queries where PRV produces incorrect answers into four failure modes. (1) *Insufficient evidence* (38.2% of errors): the evidence plan identifies the correct modalities but the retriever fails to surface the relevant segment, typically in videos exceeding 2 hours where evidence is scattered across distant temporal locations. (2) *False contradiction* (24.7%): the polarity scorer assigns contradiction to visually ambiguous evidence, leading the model to over-weight counter-evidence that is not genuinely contradictory. (3) *Planning mismatch* (21.4%): the evidence plan omits a critical modality—most commonly, queries requiring OCR evidence are misclassified as visual-only, missing on-screen text that resolves the question. (4) *Reasoning errors* (15.8%): the LLM correctly identifies and retrieves relevant evidence but synthesizes an incorrect answer, often for questions requiring multi-hop temporal reasoning spanning more than three events. Among the 7.6% of queries where PRV abstains, 78.7% would indeed have been answered incorrectly, confirming that the abstention mechanism is well-calibrated to the model's actual uncertainty.

5 Conclusion

We presented Plan-Retrieve-Verify, an LLM-driven multimodal framework that reformulates long-video audio-visual reasoning as iterative evidence acquisition with counterfactual verification, contributing three innovations. First, LLM-driven evidence planning and tool orchestration (C1) replaces passive retrieval with active evidence control, reducing tool invocations by 32–39% compared to agentic baselines while improving accuracy by 3.8–5.1 points over all compared methods. Second, the closed-loop Plan–Retrieve–Verify–Counter-Retrieve cycle (C2) enables iterative hypothesis refinement, achieving 11.6–12.9 point higher evidence recall than the best retrieval baselines. Third, counterfactual tool-use trace supervision (C3) directly trains the LLM's planning and verification decisions, reducing the average accuracy drop under modality perturbations to 4.1 points compared to 6.4–8.0 for existing methods.

Two limitations remain: the fixed step budget $S_{\max}$=8 may be suboptimal for queries of varying complexity, and the counter-retrieve mechanism occasionally flags false contradictions from visually ambiguous scenes (21.8% of counter-retrieve activations). Future work includes streaming video [5, 49] with incremental workspace updates, smaller controller LLMs for edge deployment, and user feedback for interactive evidence refinement. Cross-modal contradiction detection must be deployed responsibly with human oversight, and code and models will be released.

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